TEXORA · LECTURE SHEET

Test of Significance

Statistical hypotheses, errors and power, critical regions and p-values, the five-step testing procedure, and large-sample z-tests for a proportion and for a mean — with the standard critical values used throughout hypothesis testing.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 9 · LEVEL 2 TERM 1

1. Why we test: from a sample to a claim about a population

In estimation we use a sample to give a value, or an interval, for an unknown population parameter. In a **test of significance** we start from a definite claim about the population and ask whether the sample evidence is strong enough to reject it. A mill may claim a certain mean yarn strength, a coin may be claimed fair, or a finish may be claimed to improve a fabric. We cannot examine the whole population, so we examine a sample, allowing for the fact that samples vary by chance even when the claim is true.

The central idea is controlled risk. If the claim were true, how surprising would a result at least as extreme as the one observed be? If it would be very rare under the claim, we treat the claim itself as doubtful; if it occurs commonly by chance, we have no sufficient reason to reject the claim. A test never proves a claim absolutely; it weighs evidence against a stated standard of rarity.

2. Statistical hypotheses: the null and the alternative

A **statistical hypothesis** is a statement about a population parameter, or about the form of a population distribution, which may or may not be true. Every test sets up two competing statements.

The **null hypothesis**, written H_0, is the claim put on trial. It usually states no effect, no difference, or no change from a specified value: H_0: $p = 0.5$ for a fair coin, or H_0: $\mu = 50$ for a population mean. Under it, any difference seen in the sample is attributed to random sampling variation alone. The **alternative hypothesis**, written H_1, is what we accept if the evidence leads us to reject H_0. Its form decides whether the test is one-tailed or two-tailed, and it must be fixed before the data are examined.

FORM OF H_1	TYPE OF TEST	MEANING AND USE
H_1: $\mu \neq \mu_0$	Two-tailed	Departure in either direction matters; used when we only ask whether the parameter differs from the stated value.

$H_1: \mu > \mu_0$	One-tailed (right)	Only an increase matters; used, for example, when testing whether a treatment raises strength.
$H_1: \mu < \mu_0$	One-tailed (left)	Only a decrease matters; used, for example, when testing whether a process has reduced quality below a standard.

The choice matters. A two-tailed test at the 5% level places 2.5% of the rejection probability in each tail, whereas a one-tailed test places the whole 5% in one tail, so its critical value is smaller on that side. Choosing a one-tailed test after seeing which way the sample points would cheat the stated level, and is not permitted.

3. Level of significance, test statistic and critical region

The **level of significance**, written α, is the probability of the error we are willing to risk when we reject H_0. The conventional levels are 5% ($\alpha = 0.05$) and 1% ($\alpha = 0.01$). A 5% level means that, if H_0 is true, we shall still wrongly reject it in about 5 tests out of 100 in the long run; the stricter 1% level demands stronger evidence.

A **test statistic** is a quantity computed from the sample whose **sampling distribution** is known when H_0 is true. For large samples the standardised statistic is referred to the standard normal distribution and written z; for small samples the t, χ^2 and F distributions take its place (Topics 10 and 11). The values of the statistic divide into the **critical region** (rejection region) of values so extreme that we reject H_0 if the computed statistic falls there, and the **acceptance region** of all remaining values. The boundary values are the **critical values**, read from tables at the chosen α and number of tails.

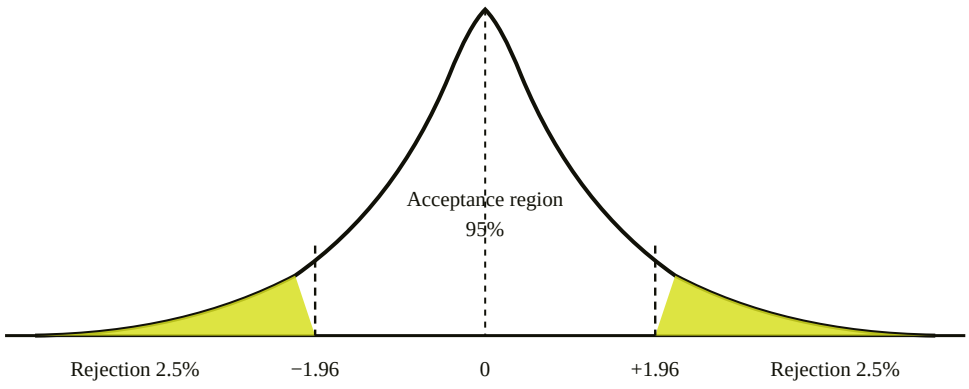


Figure 1 — Two-tailed test at the 5% level: rejection regions in the two tails beyond ±1.96, acceptance region in the centre

Figure 1 shows the picture behind every large-sample test in this topic. Under H_0 the standardised statistic follows the standard normal curve. Values beyond ±1.96 occur with total probability only 0.05,

so they form the critical region at the 5% level. A calculated |z| larger than 1.96 is therefore described as **significant at the 5% level**.

4. The two kinds of error, and power

Because the decision rests on a sample, it can be wrong in two distinct ways. Keeping the two errors separate is essential, because reducing one tends to increase the other for a fixed sample size.

TRUTH \ DECISION	ACCEPT H_0 (DO NOT REJECT)	REJECT H_0
H_0 is true	Correct decision, probability $1 - \alpha$	**Type I error**, probability α
H_0 is false	**Type II error**, probability β	Correct decision, probability $1 - \beta$

A **Type I error** is rejecting a true null hypothesis: condemning a fair coin, or a satisfactory batch, when nothing is actually wrong. Its probability is exactly α, so the experimenter controls it directly. A **Type II error** is failing to reject a false null hypothesis: passing a biased coin or an unsatisfactory batch as satisfactory. Its probability β depends on how far the true parameter lies from the value in H_0, on α, and on the sample size.

The **power** of a test is its ability to detect a false null hypothesis when it is false:

$$\text{Power} = 1 - \beta$$

Power rises when the sample size increases, when the true departure from H_0 is larger, and when α is made less strict. With a small sample a test may have low power, so failure to reject is weak evidence, not proof that H_0 is true; careful language says “do not reject H_0” rather than claiming H_0 has been proved.

5. The p-value

The **p-value** is the probability, calculated assuming H_0 is true, of obtaining a test statistic at least as extreme as the one actually observed, in the direction indicated by H_1. It measures how compatible the data are with H_0 on a continuous scale, instead of forcing an immediate yes or no answer.

The decision rule is simple: reject H_0 if $p \le \alpha$; do not reject it if $p > \alpha$. A small p-value means the result would be rare under H_0, so the evidence against H_0 is strong; a large p-value means ordinary sampling variation. The p-value is not the probability that H_0 is true; it is a probability about the data, computed assuming H_0 holds.

6. The five steps of the testing procedure

1. **State the hypotheses.** Write H_0 and H_1 in terms of the population parameter, and decide from the wording of the problem whether the test is one-tailed or two-tailed.
2. **Fix the level of significance.** Choose α, usually 5% or 1%, before computing anything from the sample.
3. **Choose the test statistic.** Select z, t, χ^2 or F according to the parameter, the sample size and what is known about the population, and note its sampling distribution and degrees of freedom where relevant.
4. **Compute the statistic.** Calculate its value from the sample, showing the standard error and every substitution, and obtain the critical value, or the p-value, for the chosen level.
5. **Decide and conclude.** Compare the computed value with the critical value (or the p-value with α), state whether H_0 is rejected, and then translate that statistical decision into a plain conclusion about the original claim.

Following the steps in this order protects the logic of the test. In particular, hypotheses and the level must not be adjusted after the calculation merely to obtain a preferred verdict.

7. Standard critical values of z

LEVEL OF SIGNIFICANCE	TWO-TAILED CRITICAL VALUE	ONE-TAILED CRITICAL VALUE
5% (α = 0.05)	1.96	1.645
1% (α = 0.01)	2.58	2.33

These four numbers should be known by heart. Note the pattern: a stricter level needs a larger critical value, and a two-tailed test needs a larger critical value than a one-tailed test at the same level, because its α is split between the two tails. The same comparison, computed value against table value, is used with t and χ^2 in the next two topics; only the table changes.

8. Worked example: a large-sample z-test of a proportion

WORKED EXAMPLE

A coin is tossed 100 times and gives 58 heads. Test, at the 5% level, whether the coin is fair.

Step 1 — Hypotheses. H_0: p = 0.5 (the coin is fair); H_1: p ≠ 0.5 (two-tailed, since bias in either

direction would count against fairness).

Step 2 — Level. $\alpha = 0.05$, so the two-tailed critical value is 1.96.

Step 3 — Statistic. Under H_0 the expected number of heads is $np = 100 \times 0.5 = 50$, and the standard error of the number of heads is $\sqrt{(npq)} = \sqrt{(100 \times 0.5 \times 0.5)} = 5$ heads. Equivalently, for the proportion, $SE = \sqrt{(0.5 \times 0.5 / 100)} = 0.05$.

Step 4 — Computation. $z = (58 - 50) / 5 = 1.60$. In proportion form, $\hat{p} = 0.58$ and $z = (0.58 - 0.50) / 0.05 = 1.6$, the same value.

Step 5 — Decision. $|z| = 1.60 < 1.96$, so the statistic does not fall in the critical region. Accept H_0 at the 5% level: 58 heads in 100 tosses is within ordinary chance variation, and there is no evidence that the coin is biased.

Eight heads above expectation sounds like a lot, but the standard error is 5 heads, so the excess is only 1.60 standard errors. Significance is always judged relative to the sampling variation, never from the raw difference alone.

9. Worked example: a z-test for a mean, and why the level matters

WORKED EXAMPLE

A random sample of $n = 64$ observations has mean $\bar{x} = 52$. The population standard deviation is known to be $\sigma = 8$. Test the claim that the population mean is $\mu = 50$.

The standard error of the mean is $\sigma / \sqrt{n} = 8 / \sqrt{64} = 8 / 8 = 1$. Hence $z = (\bar{x} - \mu) / (\sigma / \sqrt{n}) = (52 - 50) / (8 / 8) = 2.0$.

At the 5% level the two-tailed critical value is 1.96, and $2.0 > 1.96$, so H_0 is **rejected** at the 5% level. At the 1% level the critical value is 2.58, and $2.0 < 2.58$, so H_0 is **accepted** (not rejected) at the 1% level.

There is no contradiction: the same evidence meets the 5% standard but not the stricter 1% standard, which demands more protection against a Type I error. A complete report always names the level — "significant at 5% but not at 1%" is a precise and common conclusion — and the level and tails must be fixed in advance, since either can change the verdict for a result near the critical value.

10. Significance tests and estimation; large and small samples

Estimation and testing answer different questions about the same parameter. Estimation asks "what value, or range of values, is plausible?" and answers with a point estimate or a confidence interval. Testing asks "is one specified value tenable in the light of the sample?" and answers with a decision at a

stated level. Roughly, the values inside a 95% confidence interval are those a two-tailed 5% test would not reject; a good analysis often reports both, the interval showing the size of an effect and the test its statistical standing.

SITUATION	TEST	WHERE TREATED
Large sample, mean or proportion, σ known or well estimated	z-test (normal)	This topic
Small sample, mean, σ unknown, normal population	t-test	Topic 10
Frequencies, goodness of fit, independence in a table	χ^2-test	Topic 11
Comparing two or more variances or means across groups	F-test	Small-sample theory beyond this sheet

For large samples (conventionally n above about 30) the central limit theorem makes the normal approximation dependable and s is a reliable substitute for σ, so z procedures are used. For small samples that approximation is unsafe, and the exact distributions — t for means, χ^2 for frequencies and variances — take its place. The logic of Sections 2 to 6 is identical in every case: hypotheses, level, statistic, computation, decision.

Glossary

TERM	MEANING
Null hypothesis (H_0)	The claim put on trial, usually stating no effect, no difference, or a specified parameter value.
Alternative hypothesis (H_1)	The statement accepted if H_0 is rejected; its form fixes whether the test is one-tailed or two-tailed.
Level of significance (α)	The probability of a Type I error that the experimenter is willing to risk, commonly 0.05 or 0.01.
Test statistic	A quantity computed from the sample whose distribution under H_0 is known.

Critical region	The set of values of the test statistic that lead to rejection of H_0.
Type I error	Rejecting H_0 when H_0 is true; its probability is α.
Type II error	Failing to reject H_0 when H_0 is false; its probability is β.
Power	The probability of correctly rejecting a false H_0, equal to $1 - \beta$.
p-value	The probability under H_0 of a result at least as extreme as the one observed.

Key points & summary

- A test weighs sample evidence against a null hypothesis H_0; the alternative H_1 decides whether the test is one-tailed or two-tailed, and must be fixed in advance.
- The level of significance α is the probability of a Type I error; the usual levels are 5% and 1%.
- A Type I error rejects a true H_0 (probability α); a Type II error fails to reject a false H_0 (probability β); power $= 1 - \beta$.
- Reject H_0 when the computed statistic falls in the critical region, equivalently when $p \le \alpha$.
- The five steps are: state hypotheses, fix α, choose the statistic, compute it, then decide and conclude in plain language.
- Critical values of z: 1.96 and 1.645 at 5% (two-tailed and one-tailed), 2.58 and 2.33 at 1%.
- A result may be significant at 5% but not at 1%; always report the level, and read “accept H_0” as “insufficient evidence to reject”, not as proof.

Practice

Answer in your own words; brief model answers follow.

1. State, with symbols, a suitable null and alternative hypothesis for testing whether a new dyeing process changes the mean colour fastness of a fabric, and say whether the test is one-tailed or two-tailed.
2. Explain the difference between a Type I and a Type II error in the context of testing whether a batch of yarn meets a strength claim, and state the probability of each.
3. A test is carried out at the 1% level instead of the 5% level, with the same sample. What happens to the critical value, to the risk of a Type I error, and to the power of the test?
4. A coin is tossed 100 times and gives 62 heads. Using SE = 5 heads, compute z and state the decision at the 5% level about fairness.

5. What is a p-value, and what decision follows at the 5% level if the p-value of a test is 0.03? If it is 0.20?
6. Why is it misleading to say that accepting H_0 proves that H_0 is true, particularly when the sample is small?

Answer key

1. H_0: $\mu = \mu_0$, the process leaves mean fastness unchanged at its standard value; H_1: $\mu \neq \mu_0$, since a change in either direction matters. The test is two-tailed.
2. A Type I error rejects a batch that truly meets the claim (probability α, the chosen level); a Type II error passes a batch that truly falls short (probability β). The first wastes good material, the second lets weak yarn go forward.
3. The critical value increases (for z, from 1.96 to 2.58 two-tailed), the Type I risk falls from 0.05 to 0.01, and the power falls because a false H_0 is now harder to reject with the same sample.
4. $z = (62 - 50) / 5 = 2.40$. Since $2.40 > 1.96$, reject H_0 at the 5% level: the result is significant evidence against fairness.
5. The p-value is the probability under H_0 of a result at least as extreme as the one observed. $p = 0.03 \leq 0.05$, so reject H_0 at the 5% level; $p = 0.20 > 0.05$, so do not reject H_0.
6. Failure to reject only means the sample did not provide strong enough evidence against H_0. A small sample gives a low-power test with a large β, so a false H_0 can easily pass undetected; absence of evidence is not evidence of absence.

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